Messziele-Referenz — Endpunkte zum Nachpingen

Vantage Point: AWS EC2 eu-central-1 (Frankfurt) · IPs aus DNS-Auflösung 2026-06-14, per Layer 1 (ASN/RTT) bestätigt · CDN-/Round-Robin-Hosts ggf. live anders auflösen · **keine API-Keys enthalten**.

Kat. · Anbieter	Host	Prot.	Aufgelöste IPs (2026-06-14)	ASN / Org · RTT	Term.
STT Deepgram	api.deepgram.com	WSS	208.184.56.200 · 38.68.64.131/.132 (~6 IPs, Round-Robin)	AS6461 Zayo / AS174 Cogent (US) · ~101–148 ms	Host
STT Rev.ai	api.rev.ai	WSS	52.36.23.89 · 52.32.160.52 · 32.186.17.136	AS16509 Amazon (US) · ~140 ms	Host
STT Azure	italynorth.stt.speech.microsoft.com	WSS	4.232.100.212	AS8075 Microsoft (Italy North) · ~11 ms	Host
LLM OpenAI	api.openai.com	HTTPS+SSE	172.66.0.243 · 162.159.140.245	AS13335 Cloudflare · ~1 ms	Edge
LLM Groq	api.groq.com	HTTPS+SSE	172.64.149.20 · 104.18.38.236	AS13335 Cloudflare · ~1 ms	Edge
LLM Mistral	api.mistral.ai	HTTPS+SSE	172.66.2.203 · 162.159.142.207	AS13335 Cloudflare · ~1 ms	Edge
TTS Deepgram	api.deepgram.com	HTTPS-Stream	wie STT-Deepgram (Round-Robin, ~6 IPs)	AS6461 Zayo / AS174 Cogent (US) · ~140 ms	Host
TTS OpenAI	api.openai.com	HTTPS-Stream	172.66.0.243 · 162.159.140.245	AS13335 Cloudflare · ~1 ms	Edge
TTS Azure	italynorth.tts.speech.microsoft.com	HTTPS-Stream	4.232.100.220	AS8075 Microsoft (Italy North) · ~11 ms	Host

Volle URLs & Auth (Platzhalter <KEY>)

```
STT Deepgram — wss://api.deepgram.com/v1/listen?model=nova-3&language=en&encoding=linear16&sample_rate=16000&interim_results=true
   Auth: Header  Authorization: Token <KEY>
STT Rev.ai   — wss://api.rev.ai/speechtotext/v1/stream?content_type=audio/x-raw;rate=16000;format=S16LE;channels=1
   Auth: Query   &access_token=<KEY>
STT Azure    — wss://italynorth.stt.speech.microsoft.com/speech/recognition/conversation/cognitiveservices/v1
   Auth: Header  Ocp-Apim-Subscription-Key: <KEY>
LLM OpenAI   — https://api.openai.com/v1/chat/completions    ·  gpt-4o-mini-2024-07-18 · stream · max_tokens=50
LLM Groq     — https://api.groq.com/openai/v1/chat/completions   ·  llama-3.1-8b-instant
LLM Mistral  — https://api.mistral.ai/v1/chat/completions    ·  mistral-small-2603
   Auth (alle LLM): Header  Authorization: Bearer <KEY>
TTS Deepgram — https://api.deepgram.com/v1/speak?model=aura-2-asteria-en&encoding=mp3    ·  Header  Authorization: Token <KEY>
TTS OpenAI   — https://api.openai.com/v1/audio/speech    ·  tts-1 · voice=alloy · mp3    ·  Header  Authorization: Bearer <KEY>
TTS Azure    — https://italynorth.tts.speech.microsoft.com/cognitiveservices/v1    ·  SSML en-US-JennyNeural · Header  Ocp-Apim-Subscription-Key: <KEY>
```

Selbst nachmessen (DNS-Resolver: 8.8.8.8 · 1.1.1.1 · 9.9.9.9)

```
# DNS live auflösen (Round-Robin sichtbar)
dig <host> A +short            # oder: dig @1.1.1.1 <host> A +short

# ICMP-Ping
ping -c 5 <host>

# TCP-Ping auf Port 443 (eine RTT, genau wie in der Studie)
python3 - <<'PY'
import socket, time
h = "<host>"; ip = socket.gethostbyname(h)
```

```
t = time.perf_counter(); socket.create_connection((ip, 443), timeout=5).close()
print(round((time.perf_counter() - t) * 1000, 2), "ms ->", ip)
PY

# Route + ob sie im CDN-AS endet (TCP-Traceroute, braucht sudo)
sudo traceroute -T -p 443 -n <host>

# TLS-Version / Zertifikat
openssl s_client -connect <host>:443 -servername <host> </dev/null 2>/dev/null | openssl x509 -noout -subject

# ASN einer IP (Team-Cymru)
dig +short $(echo <ip> | awk -F. '{print $4"."$3"."$2"."$1}').origin.asn.cymru.com TXT
```

Hinweis: Aus Frankfurt sollten OpenAI/Groq/Mistral (+ OpenAI-TTS) ~1 ms zeigen (Cloudflare-Edge), Azure ~11 ms (EU-RZ), Deepgram/Rev.ai ~100–145 ms (US). Andere Standorte/Resolver → andere IPs/RTTs.